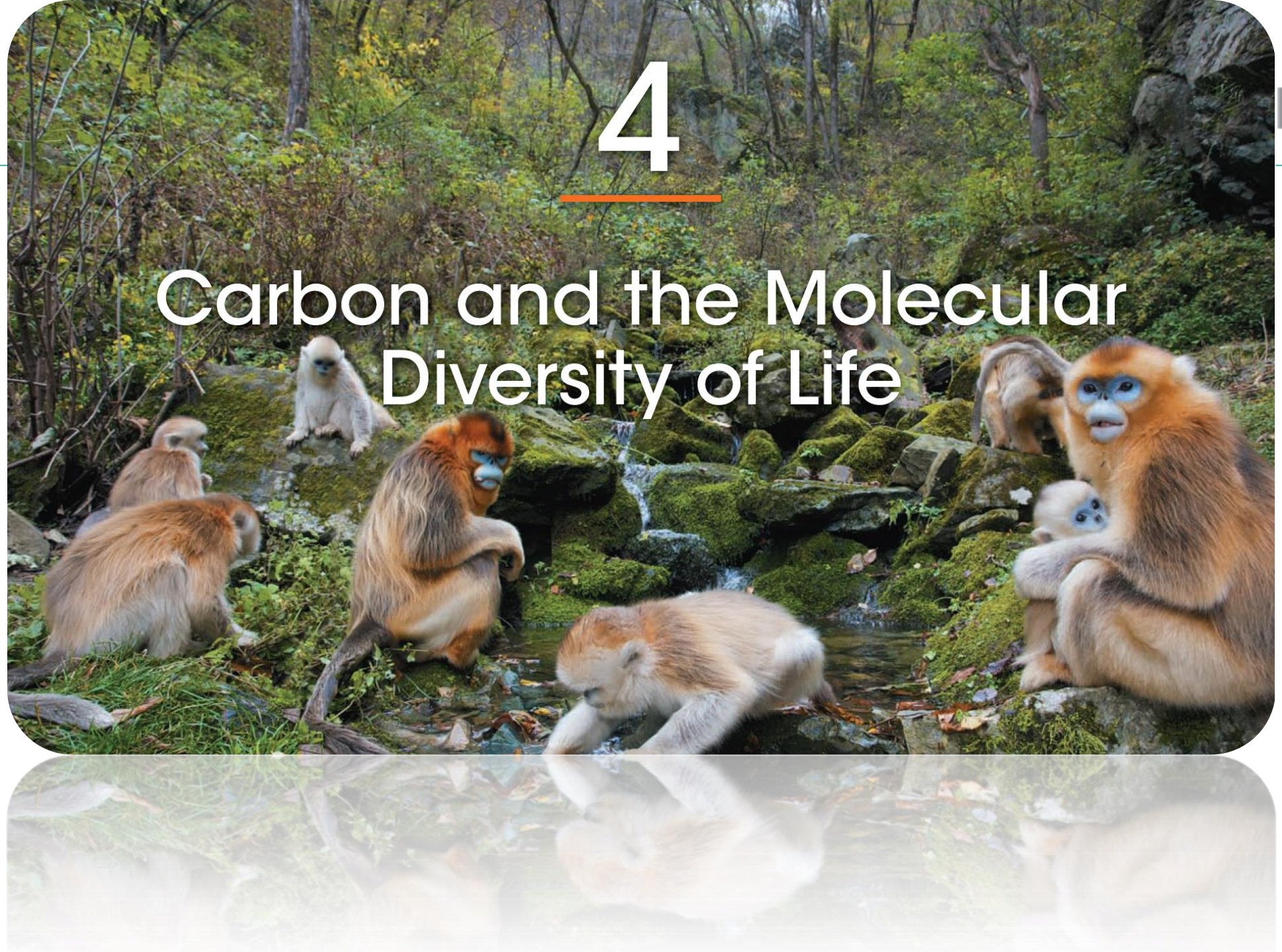
LECTURE-4

Molecular diversity

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

4

Carbon and the Molecular Diversity of Life



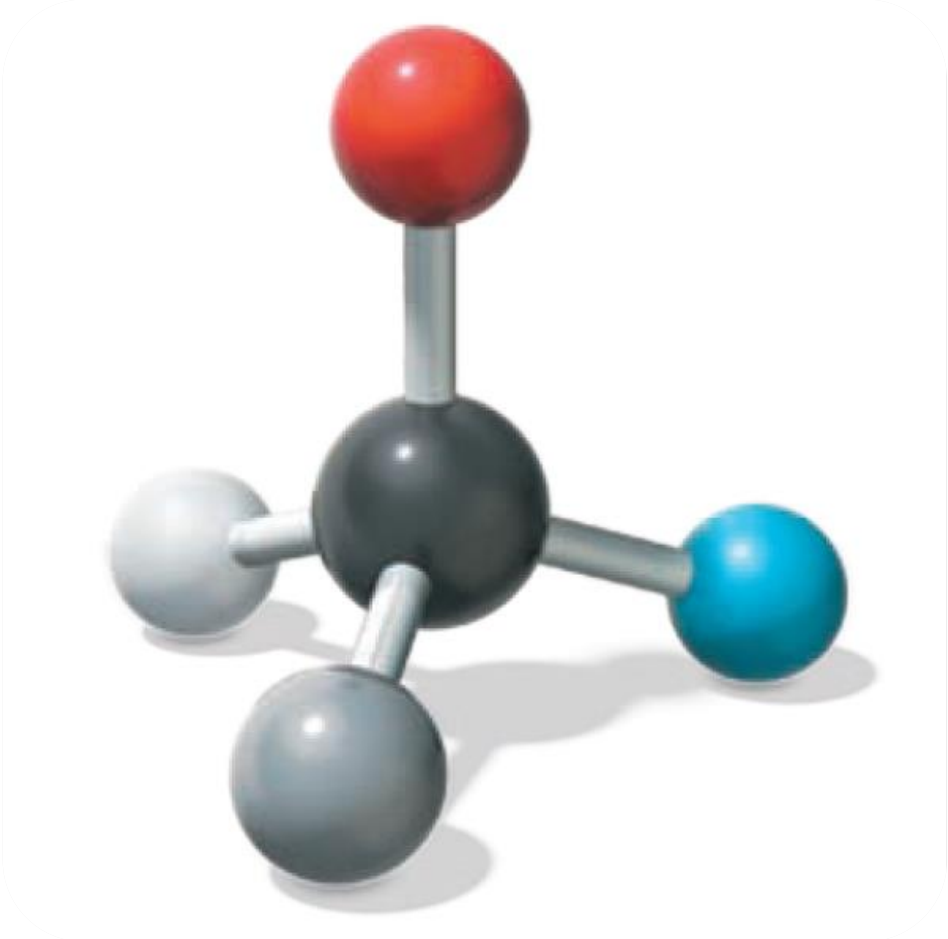
Key concepts

KEY CONCEPTS

4.1 Organic chemistry is the study of carbon compounds

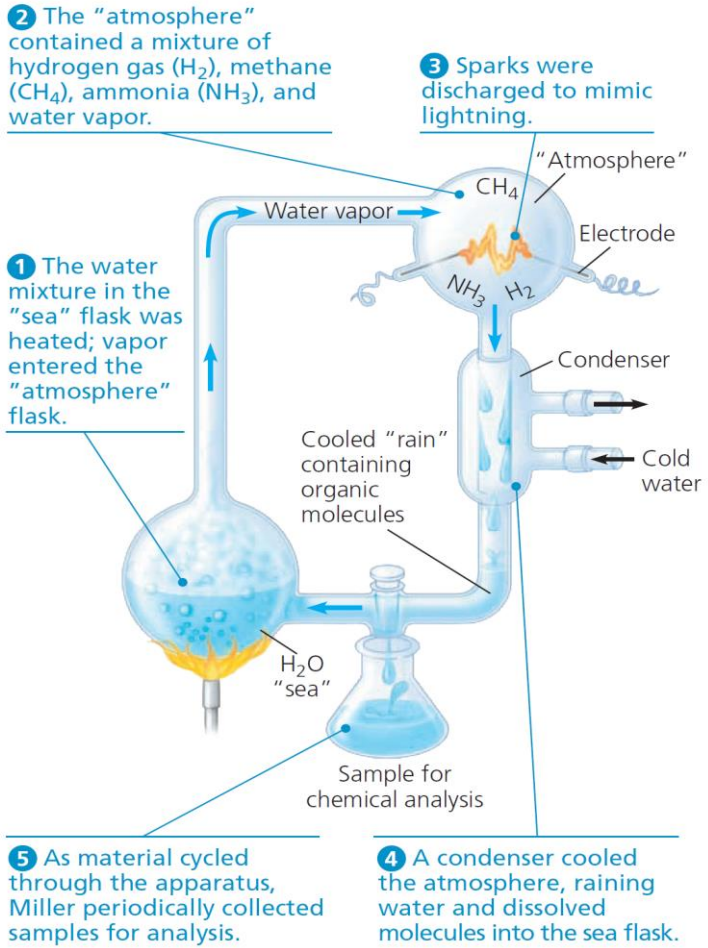
4.2 Carbon atoms can form diverse molecules by bonding to four other atoms

4.3 A few chemical groups are key to molecular function



Organic chemistry–study of life

Experiment In 1953, Stanley Miller set up a closed system to mimic conditions thought at that time to have existed on the early Earth. A flask of water simulated the primeval sea. The water was heated so that some vaporized and moved into a second, higher flask containing the "atmosphere"—a mixture of gases. Sparks were discharged in the synthetic atmosphere to mimic lightning.

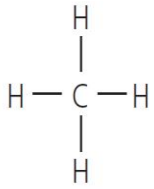
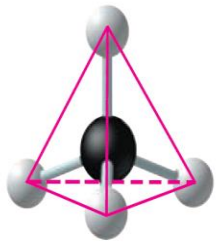
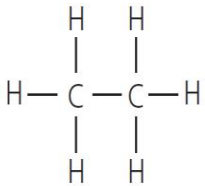
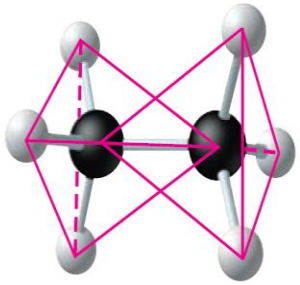
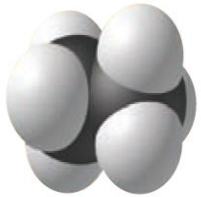
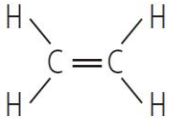
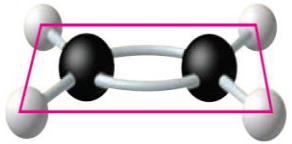


Results Miller identified a variety of organic molecules that are common in organisms. These included simple compounds, such as formaldehyde (CH_2O) and hydrogen cyanide (HCN), and more complex molecules, such as amino acids and long chains of carbon and hydrogen known as hydrocarbons.

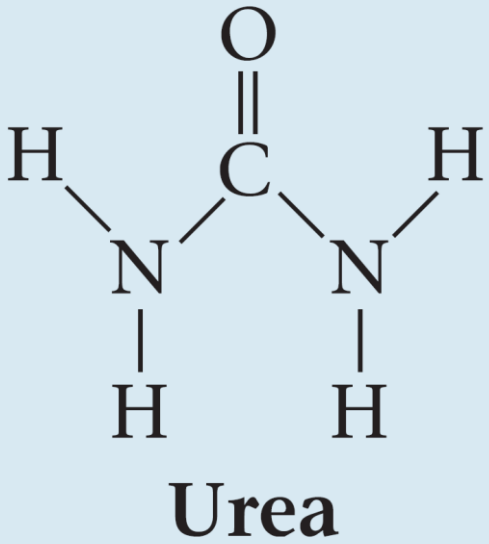
Conclusion Organic molecules, a first step in the origin of life, may have been synthesized abiotically on the early Earth. Although new evidence indicates that the early Earth's atmosphere was different from the "atmosphere" used by Miller in this experiment, recent experiments using the revised list of chemicals also produced organic molecules. (We will explore this hypothesis in more detail in Chapter 25.)

Source: S. L. Miller, A production of amino acids under possible primitive Earth conditions, *Science* 117:528–529 (1953).

Molecules and shapes

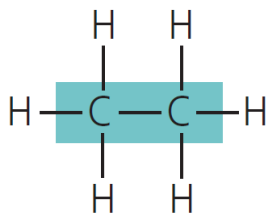
Molecule and Molecular Shape	Molecular Formula	Structural Formula	Ball-and-Stick Model (molecular shape in pink)	Space-Filling Model
(a) Methane. When a carbon atom has four single bonds to other atoms, the molecule is tetrahedral.	CH_4			
(b) Ethane. A molecule may have more than one tetrahedral group of single-bonded atoms. (Ethane consists of two such groups.)	C_2H_6			
(c) Ethene (ethylene). When two carbon atoms are joined by a double bond, all atoms attached to those carbons are in the same plane, and the molecule is flat.	C_2H_4			

- Urea and carbon dioxide are molecules with only onecarbon atom.

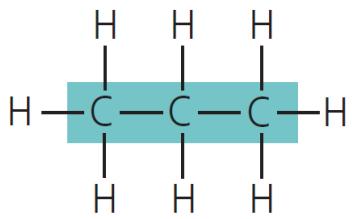


Molecular diversity

(a) Length



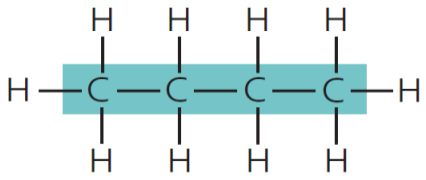
Ethane



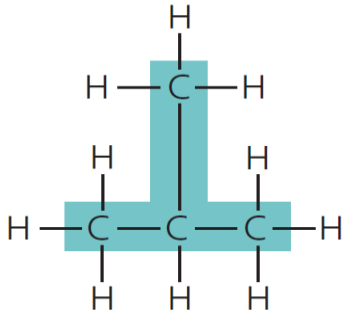
Propane

Carbon skeletons vary in length.

(b) Branching



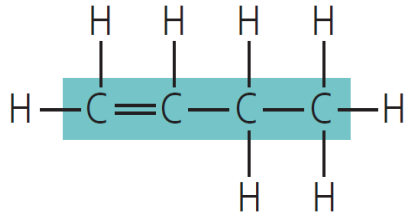
Butane



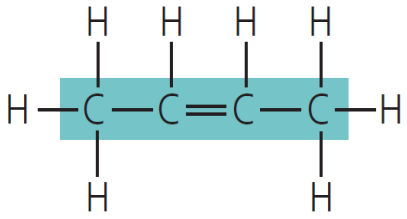
2-Methylpropane
(commonly called isobutane)

Skeletons may be unbranched or branched.

(c) Double bond position



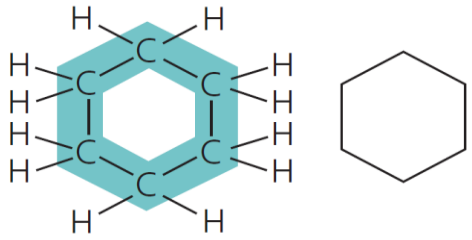
1-Butene



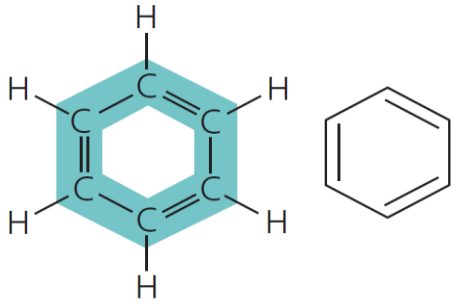
2-Butene

The skeleton may have double bonds, which can vary in location.

(d) Presence of rings



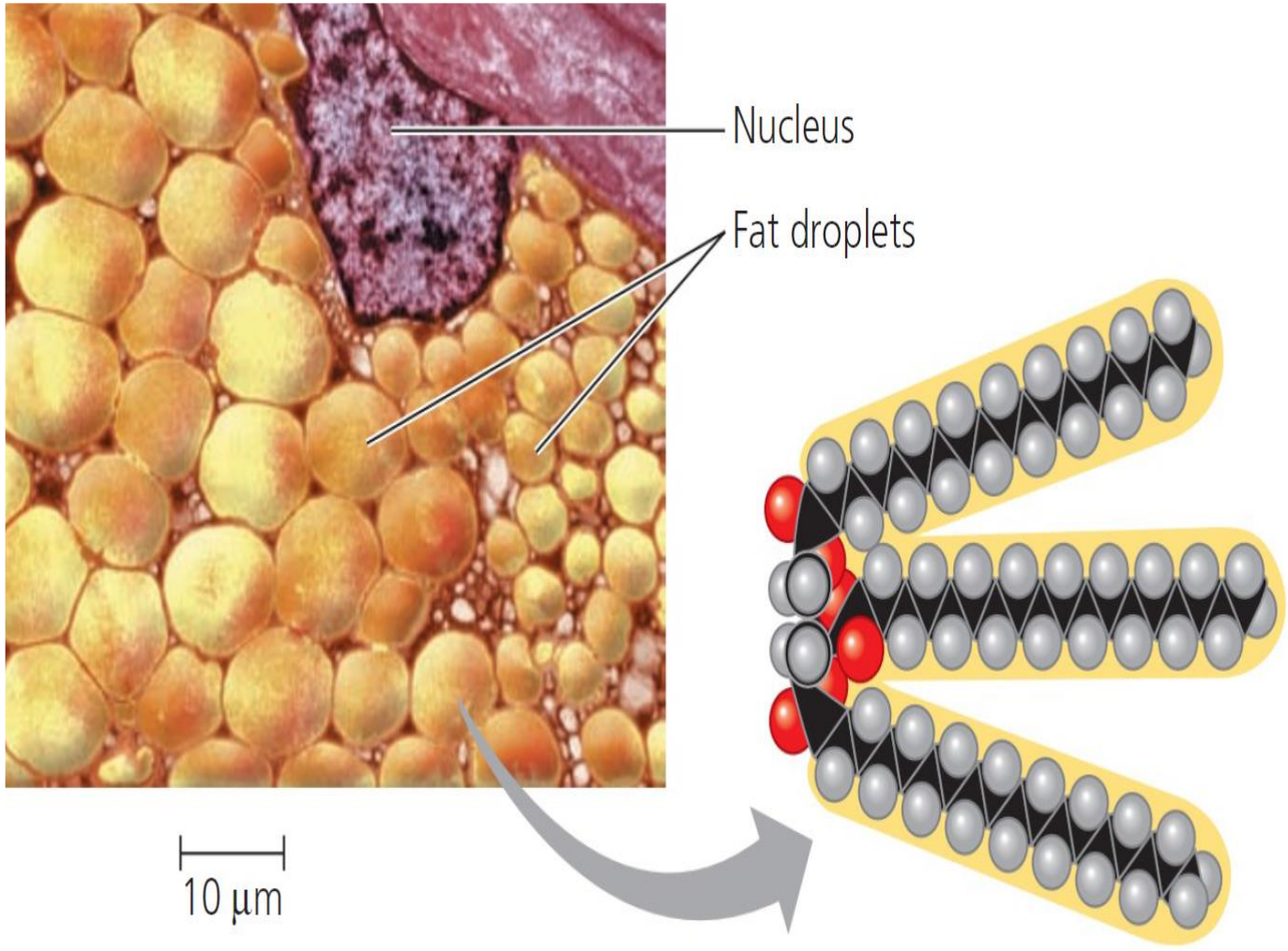
Cyclohexane



Benzene

Some carbon skeletons are arranged in rings. In the abbreviated structural formula for each compound (at the right), each corner represents a carbon and its attached hydrogens.

Hydrophobic behavior of fat

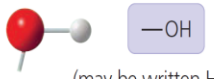
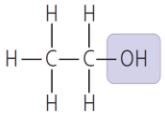
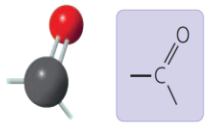
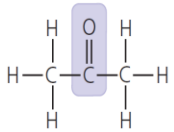
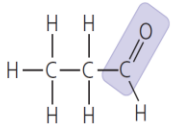
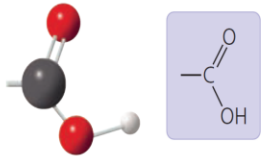
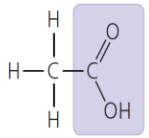
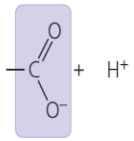
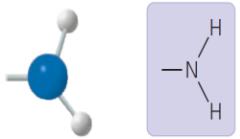
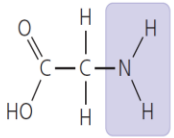
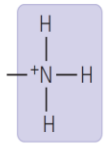


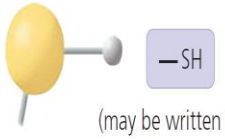
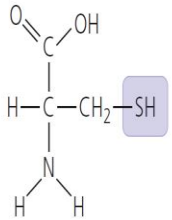
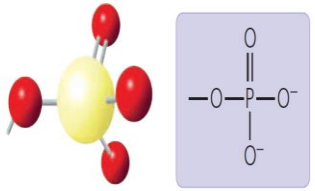
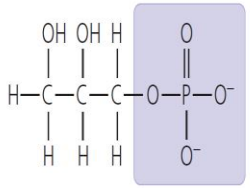
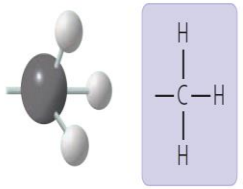
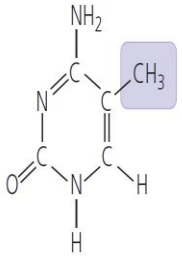
(a) Part of a human adipose cell (b) A fat molecule

(a) Mammalian adipose cells stockpile fat molecules as a fuel reserve. This colorized micrograph shows part of a human adipose cell with many fat droplets, each containing a large number of fat molecules.

(b) A fat molecule consists of a small, nonhydrocarbon component joined to three hydrocarbon tails that account for the hydrophobic behavior of fats. The tails can be broken down to provide energy. (Black = carbon; gray = hydrogen; red = oxygen.)

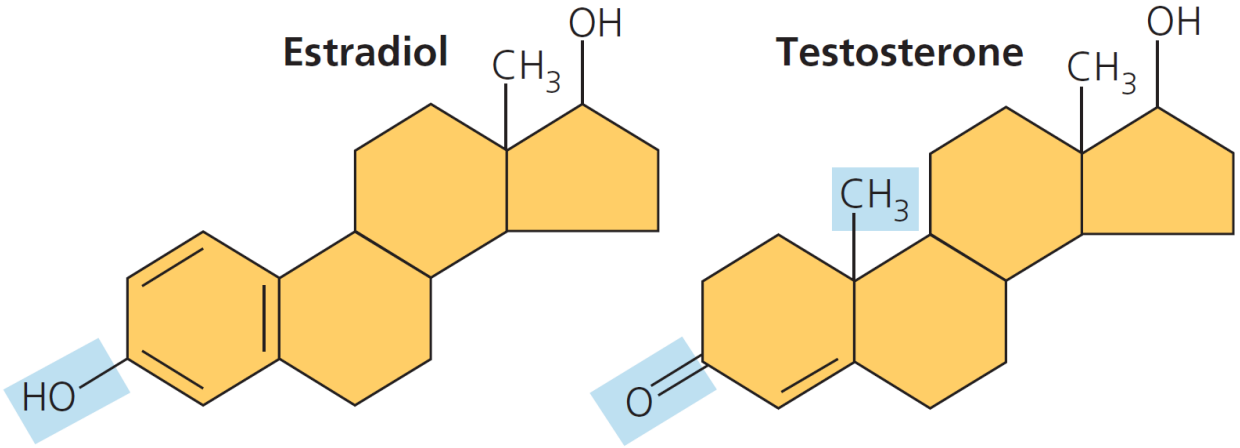
A few chemical groups are key to molecular function

Chemical Group	Group Properties and Compound Name	Examples
Hydroxyl group (—OH)  (may be written HO—)	Is polar due to electronegative oxygen. Forms hydrogen bonds with water, helping dissolve compounds such as sugars. Compound name: Alcohol (specific name usually ends in *-ol*)	 **Ethanol**, the alcohol present in alcoholic beverages
Carbonyl group (>C=O) 	Sugars with ketone groups are called ketoses; those with aldehydes are called aldoses. Compound name: Ketone (carbonyl group is within a carbon skeleton) or aldehyde (carbonyl group is at the end of a carbon skeleton)	 **Acetone**, the simplest ketone  **Propanal**, an aldehyde
Carboxyl group (—COOH) 	Acts as an acid (can donate H^+) because the covalent bond between oxygen and hydrogen is so polar. Compound name: Carboxylic acid, or organic acid	 **Acetic acid**, which gives vinegar its sour taste ⇌  Ionized form of —COOH (carboxylate ion), found in cells
Amino group ($—NH_2$) 	Acts as a base; can pick up an H^+ from the surrounding solution (water, in living organisms). Compound name: Amine	 **Glycine**, an amino acid (note its carboxyl group) ⇌  Ionized form of $—NH_2$, found in cells

Chemical Group	Group Properties and Compound Name	Examples
Sulfhydryl group (—SH)  (may be written HS—)	Two —SH groups can react, forming a "cross-link" that helps stabilize protein structure. Hair protein cross-links maintain the straightness or curliness of hair; in hair salons, permanent treatments break cross-links, then re-form them while the hair is in the desired shape. Compound name: Thiol	 **Cysteine**, a sulfur-containing amino acid
Phosphate group ($—OPO_3^{2-}$) 	Contributes negative charge (1– when positioned inside a chain of phosphates; 2– when at the end). When attached, confers on a molecule the ability to react with water, releasing energy. Compound name: Organic phosphate	 **Glycerol phosphate**, which takes part in many important chemical reactions in cells
Methyl group ($—CH_3$) 	Affects the expression of genes when on DNA or on proteins bound to DNA. Affects the shape and function of male and female sex hormones. Compound name: Methylated compound	 **5-Methyl cytosine**, a component of DNA that has been modified by addition of a methyl group

A few chemical groups are key to molecular function.

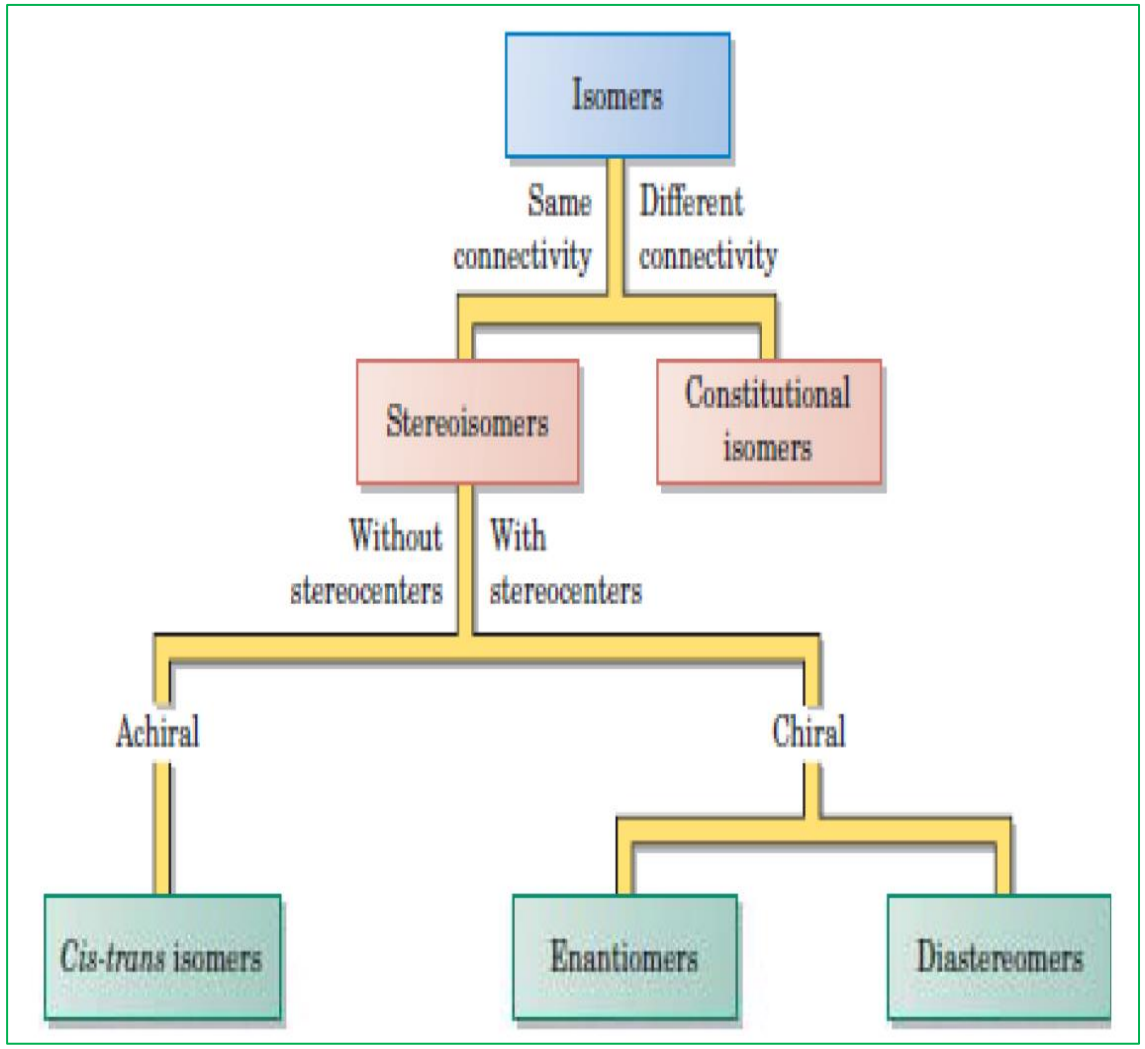
- **Hydrocarbons**, the simplest organic molecules, **are the underlying framework for more complex organic molecules**.
- Several **chemical groups can replace one or more hydrogens** of the hydrocarbon.
- Chemical groups may participate in reactions or may **contribute to function** indirectly by their effects on the molecular shape, giving each molecule its unique properties.



Female and male sex hormones in humans and other vertebrates.

Stereochemistry and isomers

- A carbon-containing compound commonly exists as stereoisomers, molecules with the same chemical bonds but different stereochemistry – i.e., compounds having a fixed spatial arrangement of atoms.
- Interactions between biomolecules are invariably stereospecific, requiring specific stereochemistry in the interacting molecules.
- Two important isomers that are often encountered are configurational and conformational.
- **Configuration** is conferred by the presence of either (1) double bonds, around which there is no freedom of rotation, or (2) chiral centers, around which substituent groups are arranged in a specific sequence.



Enantiomers, diastereomers and racemic mixtures

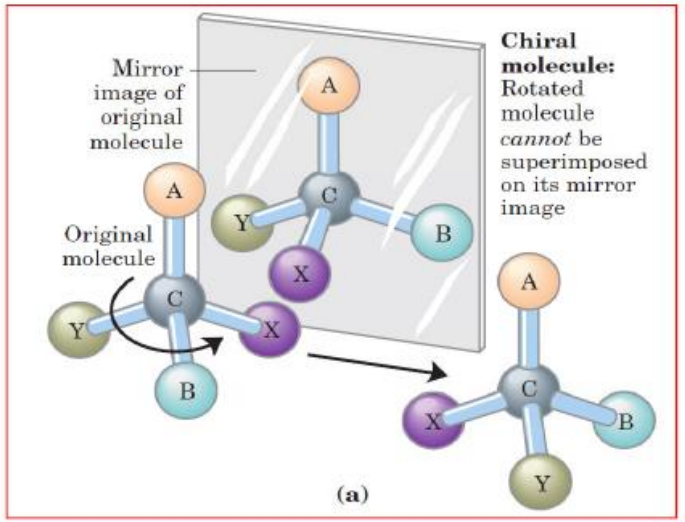
- A carbon atom with four different substituents is said to be asymmetric, and asymmetric carbons are called **chiral** centers.

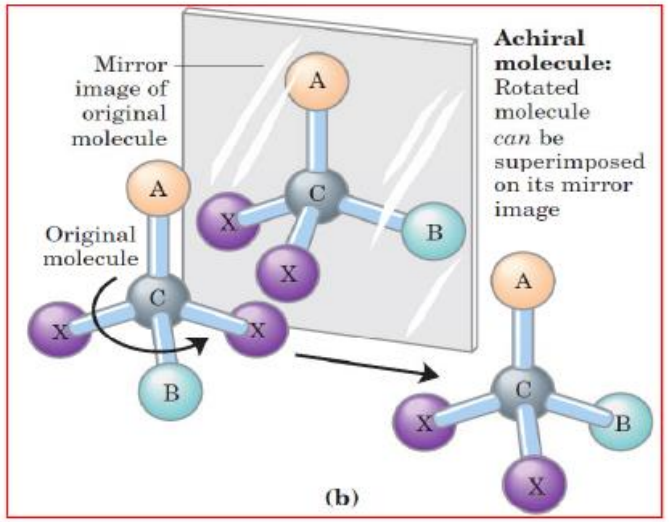
- A molecule with only one chiral carbon can have two stereoisomers; when two or more (n) chiral carbons are present, there can be 2^n stereoisomers (not always true).

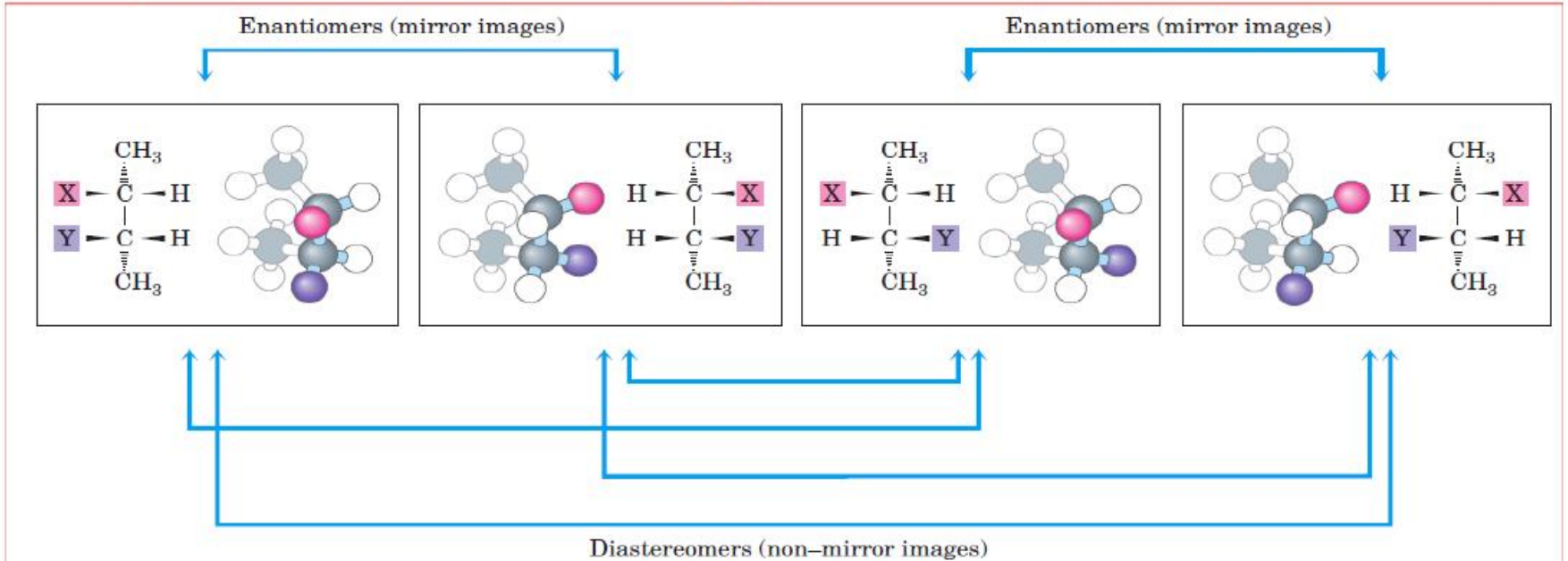
- Stereoisomers that are mirror images of each other; they are called **enantiomers**.

- Enantiomers that are not mirror images of each other are called **diastereomers** (has more than one chiral carbon).

Chiral molecules- Examples of enantiomers and diasteromers



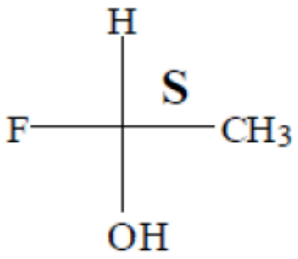


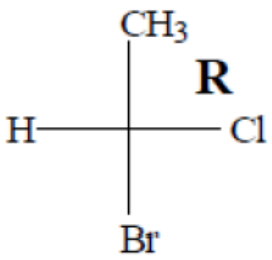


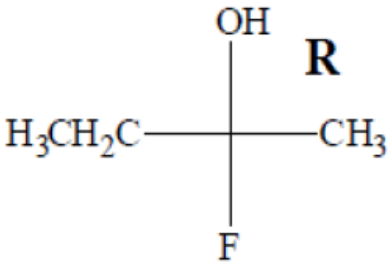
R–S nomenclature to study biomolecules

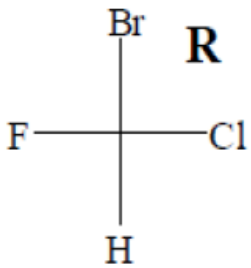
- Enantiomers have nearly identical chemical properties but differ in a characteristic physical property, their interaction with plane-polarized light.
- An equimolar solution of the two enantiomers that do not show no optical rotation is called **racemic mixture**. Compounds without chiral centers also do not rotate the plane of plane-polarized light.
- For compounds with one or more than one chiral center, the most useful system of nomenclature is the R-S system.
- In this nomenclature, each group attached to a chiral carbon is assigned a priority: $-OCH_2 > -OH > -NH2 > -COOH > CHO > -CH_2OH > -CH3 > -H$.
- For naming in the RS system, the chiral atom is viewed with the group of lowest priority pointing away from the viewer.
- If the priority of the other three groups (1 to 3) decreases in clockwise order, the configuration is (R).
- If in counterclockwise order, the configuration is (S).
- The inclusion of these designations in the name of the compound provides an unambiguous description of the stereochemistry at each chiral center.

Specification of configuration – R and S







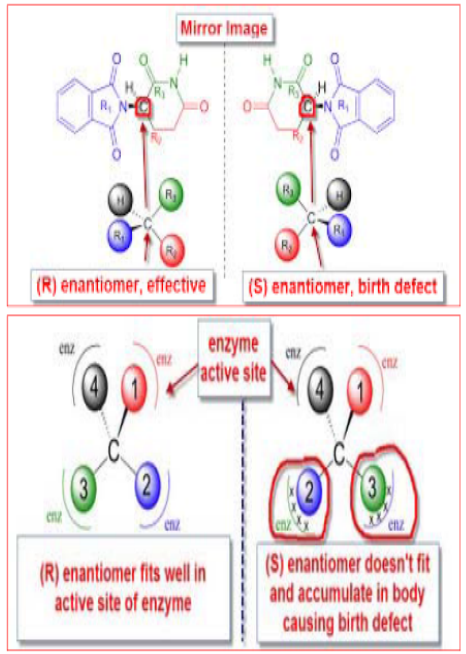


- Lowest priority number should be in the vertical axis.
- If viewing highest priority next to the next highest priority clockwise, its R configuration.
- Any molecule, first, convert to Fischer projection.
- Assign priority numbers largest atomic number will get the highest priority.
- In this course, only carbon, hydrogen, oxygen, nitrogen and halogens will be considered. Atomic numbers are C = 6, N=7, O= 8, F = 9, Cl = 17, Br = 35, I = 53.

Examples of enanatiomeric drugs

Drug	Effects	Effective Enantiomer	Ineffective Enantiomer
Ibuprofen	Reduces inflammation and pain	*S*-Ibuprofen	*R*-Ibuprofen
Albuterol	Relaxes bronchial (airway) muscles, improving airflow in asthma patients	*R*-Albuterol	*S*-Albuterol

▲ **Figure 4.8 The pharmacological importance of enantiomers.** Ibuprofen and albuterol are drugs whose enantiomers have different effects. (*S* and *R* are used here to distinguish between enantiomers.) Ibuprofen is commonly sold as a mixture of the two enantiomers; the *S* enantiomer is 100 times more effective than the *R* form. Albuterol is synthesized and sold only as the *R* form of the drug; the *S* form counteracts the active *R* form.

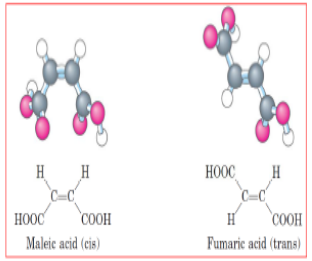


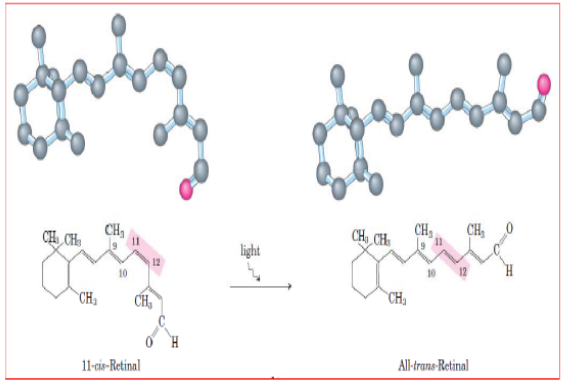
- Thalidomide drug exist as optical isomers, Enantiomers (R) and (S). Sold in Europe and in 1957, and sold under 37 brand names in 47 countries.

- Sold as Racemic mixture, but very unstable. It can be converted from one isomer to another.
- (R) effective against insomnia and morning sickness.
- (S) teratogenic, birth and limb defect.
- Banned in 1961.

Geometric isomerism

- Configurational isomers cannot be interconverted without temporarily breaking one or more single/double bonds (i.e., single C-C. bonds).
- **<u>Geometric isomer</u>** is a configuration isomer that arises due to the presence of double bound and contains no chiral carbon.
- Geometric isomers exists in Cis or trans configurations (or) E (Entgagen) or Z (Zusamann) forms.
- In geometric isomer, SP^2 hybridized double π bonds are broken so that free rotation about C-C is possible.
- The free rotation followed by SP^2 π bond formation results in different configuration (either cis (Z) or trans (E) forms).





- In the vertebrate retina, the initial event in light detection is the absorption of visible light by 11-cis-retinal.

- The energy ofthe absorbed light (about 250 kJ/mol) converts 11-cis-retinal to all-trans-retinal, triggering electrical changes in the retinal cell that lead to a nerve impulse.
- What is the arrow mechanism) that can be plausibly given for this conversion?